

Explanatory and Response Variables

MATH 213

Explanatory and Response

If we are using one variable to help us understand or predict values of another variable, we call the former the *explanatory variable* and the latter the *response variable*

Examples:

- Does just one session of cognitive behavioral therapy improve sleep?
- Does the use of stents reduce the incidence of stroke?
- Does gamification improve learning outcomes for statistics students?

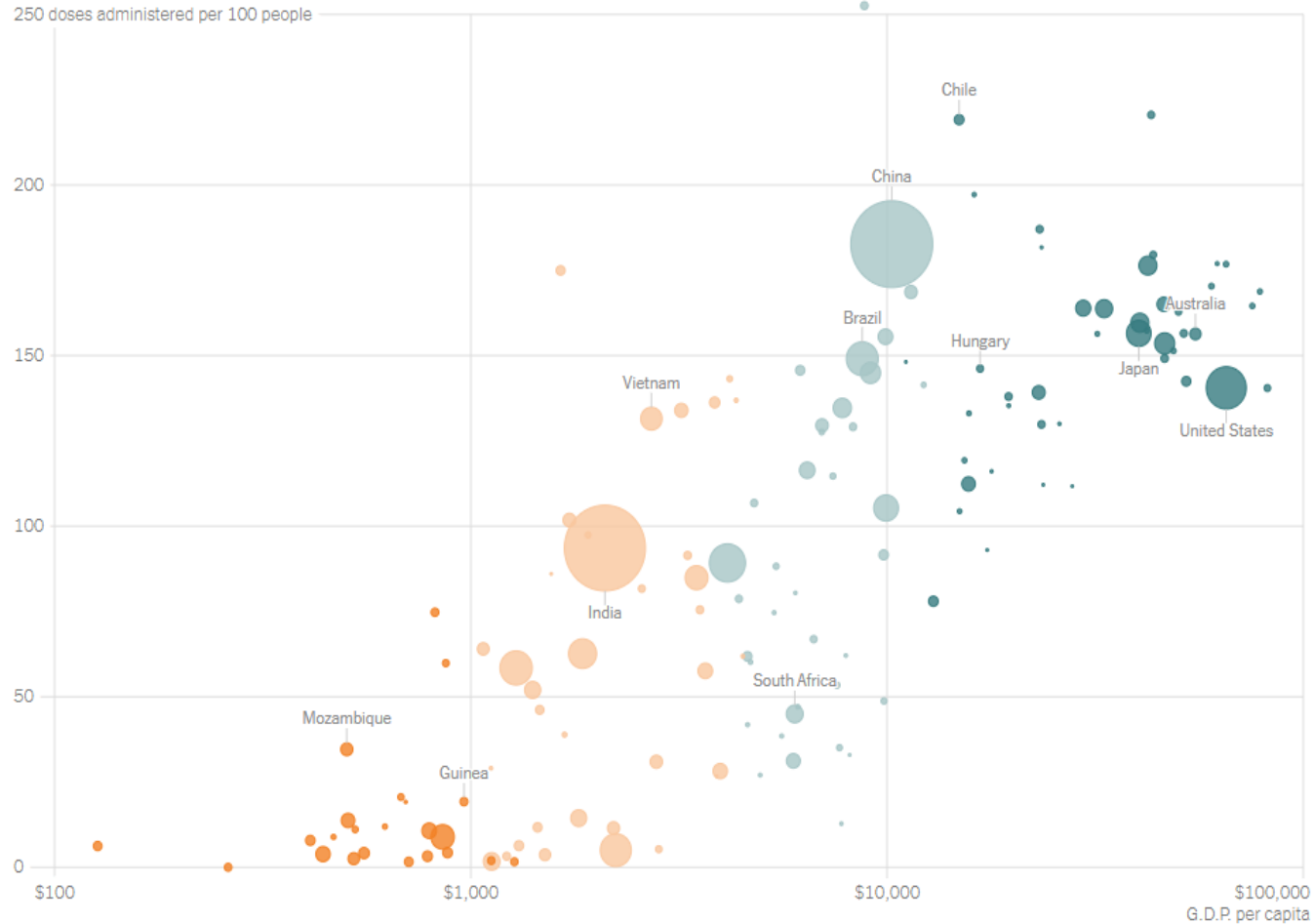
Example

Wealthier countries' head start on vaccination has kept them in the lead

Circles are sized by country population.

Income group Low Lower middle Upper middle High

► Dec. 8, 2021



Sources: Vaccination data from local governments via Our World in Data; income classifications and gross domestic product data from the World Bank. * Figures marked with an asterisk * were last reported more than two weeks ago.

Variable on x-axis?

Variable on y-axis?

Which are the researchers thinking of as being the explanatory variable? The response variable?